

Master Final Project

Implementation and evaluation of a deep
spatio-temporal scanpath prediction model for 360°
video

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ACKNOWLEDGEMENTS

I would like to express my most sincere gratitude to my supervisors, **Daniel** and **Edurne**, for their guidance and support during the completion of this work. Their knowledge and advice have been essential in carrying out this project.

I also thank the **University of Zaragoza** and the **Graphics and Imaging Lab** for the resources and academic environment provided, and my classmates and friends for their support and collaboration.

Finally, to my family, especially my parents and my brother, for their love and unconditional support throughout these years.

Thank you all.

Juan Lorente

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ABSTRACT

Virtual reality (VR) has gained increasing importance in recent years due to its wide range of applications, from content generation to rendering optimization. To create immersive experiences, it is essential to understand where users direct their attention. Trying to predict realistic scanpaths in a 360° video involves several challenges that increase the difficulty of this objective. It must be taken into account that, when working in a spherical context, the user does not see the entire scene at all times, and additional temporal complexity is introduced by working with video.

In this project, a scanpath prediction framework is proposed for 360° video that separates spatio-temporal attention modeling from scanpath generation. This project focuses on adapting a model designed for saliency map prediction and further developing it in order to predict realistic scanpaths.

The proposed model predicts temporal Scanpath Probability Maps (tSPMs) that capture the evolution of visual attention across video frames while explicitly accounting for spherical geometry and motion cues. Once all the tSPMs are predicted for a desired video, the scanpaths are generated using a probabilistic sampling method, on each of the predicted tSPM with a mask, to obtain the fixation point for the current frame, where the mask is a gaussian map centered on the previously predicted fixation applied to the tSPM.

The approach is evaluated in the VR-Eye-Tracking dataset and is tested for generalization in the Sports-360 dataset using multiple trajectory-based metrics. Experimental results show that the proposed method produces coherent and realistic scanpaths, outperforming the saliency-based baseline, achieving superior performance compared to the current state of the art approaches and are closer to the human baseline.

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Chapter 1

Introduction and Objectives

In recent years, virtual reality (VR) has progressed significantly, especially in the entertainment, video game, and education sectors. Virtual reality headsets allow for better immersion in 3D environments than flat monitors. Hardware continues to improve over time due to market competitiveness in the creation and design of new virtual reality headsets by companies such as Meta, Valve, or Apple. This makes VR more accessible and appealing to users, encouraging content creation for these devices, and expanding research in the field of VR in order to further improve these technologies, such as the evolution of haptic controllers or the development of augmented reality glasses the size of regular eyeglasses.

The creation of immersive and engaging VR content depends a lot on understanding what captures people's attention. Generating compelling content requires deep knowledge of human attention in virtual reality environments: What attracts people's attention the most? How do users visually explore content in virtual reality? What features have the greatest influence when directing attention? Therefore, in order to design more immersive experiences, it is necessary to deeply understand human behavior.

One of the main ways to measure visual attention is through the use of saliency maps, which are a visual representation of the areas of an image that attract people's attention. These maps are computed by aggregating gaze data from multiple observers viewing a scene, which may depend on visual features (such as color, contrast, and orientation) and contextual features (such as familiarity and relevance of objects). Saliency maps are useful for identifying and predicting visual attention patterns in different types of visual content and for giving content creators a preliminary idea of the expected behavior of their users, an example of a saliency map is shown in Figure 1.1.



Figure 1.1: An example of a saliency map on the left and a scanpath on the right.

However, saliency does not take into account the temporal aspect of attention, such as the order in which a scene is explored. To address this limitation, another representation of visual attention is used: *scanpaths*.

A *scanpath* is the trajectory followed by an observer when exploring an image or video over time, and consists mainly of a sequence of fixations and saccadic movements over the scenario. Fixations are periods in which the eyes remain relatively still and focus on a specific point, while saccades are the rapid movements between fixations. Scanpath analysis provides valuable information on how users visually process information over time and which elements attract their attention the most. In Figure 1.1 is shown an example of a scanpath.

In order to predict *scanpaths*, datasets are needed that capture the eye movements of multiple observers across different videos, so as to record human behavior and subsequently develop methods that can replicate patterns of human visual attention. Eye-movement records are obtained using eye-tracking technologies that make it possible to precisely measure where and for how long observers fixate their gaze. Capturing these data is costly, which has motivated many researchers to develop models capable of predicting visual behavior without requiring such large amounts of data.

There are different approaches to predicting scanpaths in images, but in all of them, the models focus on static features of the visual content [1]. However, in 360° videos, it is crucial to consider the temporal dimension and the interactivity of the environment [2], [3], because this allows users to freely explore visual content, adding an additional layer of complexity, in which prediction methods should be able to simulate the variability and uncertainty characteristic of scanpaths.

Integrating scanpath prediction methods into videos presents an additional challenge due to the dynamic nature of the content, where the movement of objects affects visual attention, which does not occur when we have static images. This involves not only predicting the areas of highest saliency in each frame but also how these areas change and how observers can shift their focus in response to events in the video [4].

In certain applications, such as viewport prediction, it is necessary for it to operate in real time, since the goal is to reduce the resources required for rendering the scene on virtual reality headsets. This makes it necessary for the process to be optimized, even at the cost of sacrificing complexity, in order to be able to perform predictions in real time.

Therefore, developing this field of study would improve the understanding of human behavior with respect to visual attention in 360° videos, allowing content creators to design immersive experiences while taking into account visual attention of users.

1.1 Objectives of the work

The main objective of this Final Master’s Project is to design, implement, and evaluate a deep spatio-temporal framework for scanpath prediction in 360° videos. This project focuses on adapting a model designed for saliency map prediction and further developing it in order to predict realistic scanpaths.

The specific objectives of this work are the following:

- Study the existing literature on scanpath and saliency prediction in images and videos, with a particular focus on 360° content, identifying their main strengths and limitations.
- Implement a deep learning model capable of predicting scanpaths in 360° video together with a heuristic sampling strategy for generating scanpaths.
- To evaluate the proposed approach using trajectory-based metrics and to compare it against baselines and state-of-the-art approaches.
- To analyze the results, both the quantitative results from the metrics used and the qualitative results from visualizing the scanpaths, where the proposed approach is compared with respect to the current state of the art.

1.2 Planning and tools

The development of this project is structured in different phases: The project starts with studying the related work on scanpath prediction and saliency modeling. This was followed by the design and implementation of the proposed model and the heuristic sampling method. The next phase is dedicated to the evaluation and analysis of the results, comparing them with state of the art methods.

In the following links, both the results obtained in **Drive**, and the framework source code in **Github** are available to promote future extensions or evolutions of this work.

In Table 1.1 and Figure 1.2 the overall planning of the project is summarized as a table with all the hours dedicated to each part and a Gantt diagram, which shows the temporal organization and overlap of the different phases.

The following resources and libraries were used to train the deep learning model, using a NVIDIA RTX A6000 GPU thanks to the environment provided by the Graphics and Imaging Lab. For the implementation, the Python programming language was used, with libraries such as PyTorch, NumPy and TensorBoard to work with deep learning.

Project Task	Hours Spent
Previous study	79
Model design	126
Model implementation	228
Sampler implementation	74
Evaluation of results	178
Write the memory	97
Total	782

Table 1.1: Hours dedicated to each part of the project.

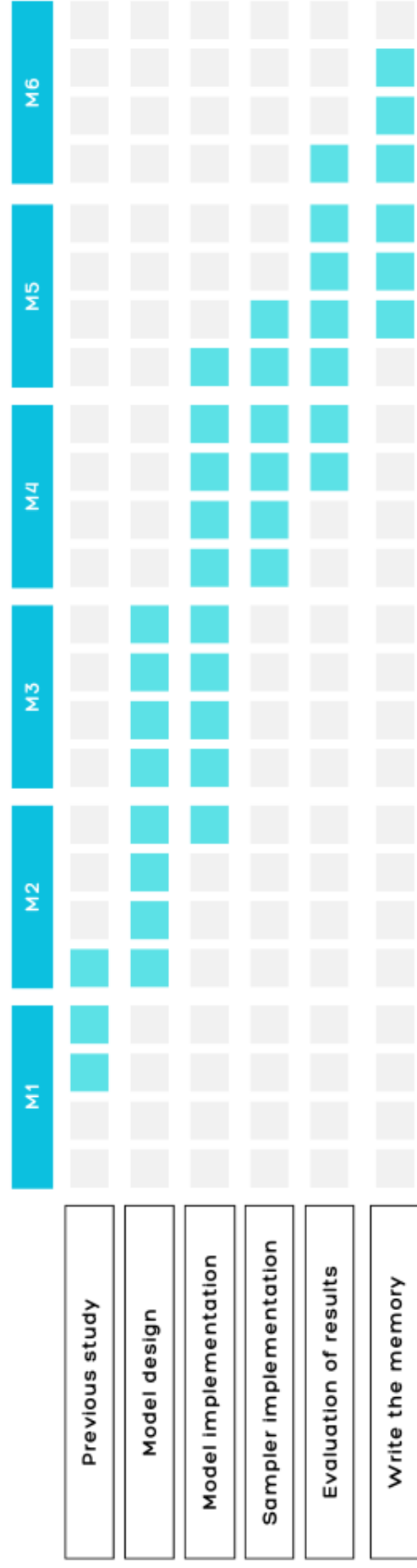


Figure 1.2: Gantt diagram.

Chapter 2

Prior Study

Due to the rise of virtual reality in recent years and the generation of content for these devices, interest in the study of human behavior in immersive environments has increased in order to advance these technologies by optimizing the resources used and improving the user experience in virtual reality.

However, predicting scanpaths for 360° videos entails different challenges in carrying out the task, such as how to model the temporal and spatial dependencies of human behavior and how to model the uncertainty and variability of human gaze.

Therefore, this prior study reviews the state of the art literature on the challenges related to this specific case study, more specifically by examining the different datasets that can be used, the different approaches to saliency prediction, and the various approaches to scanpath prediction.

In addition, at the end of the chapter the limitations of these studies are explained when applied to scanpath prediction in 360° videos, as well as how this project addresses these issues to overcome them.

2.1 Related work

2.1.1 Saliency prediction

The main approach over time for the study of human gaze was initially focused on saliency, which represents the direction toward which users direct their visual attention. In particular, the study by Itti et al. [5] laid the foundations for the prediction of visual attention in images [6, 7, 8].

Several works specifically address the challenges of 360° video by incorporating spherical geometry and motion cues into saliency prediction architectures. Like SST-Sal [9] which is a novel saliency prediction model which exploits the joint potential of spherical convolutions and recurrent neural networks (using ConvLSTMs) to extract and model the inherent spatio-temporal features.

More recent approaches extend these ideas with audiovisual integration, as proposed in AViSal360[10], an audiovisual saliency model for 360° video content, that integrates both spatialized and semantic audio information, with visual inputs to take into account how spatialized audio can affect visual attention.

Saliency maps are a good starting point from which sampling methods can be used to generate scanpaths based on the attention captured by saliency. However, these models do not take into account the dependence of attention on different regions of the scene with the previous fixation points of the scanpath.

2.1.2 Scanpath prediction

Early approaches to scanpath prediction [11, 12, 13] in 360° content include GANs and sampling-based models that learn the distributions of plausible exploration trajectories and then sample scanpaths from them.

ScanGAN360 [14], for example, generates realistic scanpaths for 360° images using spherical dynamic time warping (DTW) loss to encourage temporal similarity with human behavior. Although effective for static panoramas, this method is limited to still images, which restricts their applicability to dynamic video scenarios.

Another common line of work follows a saliency to scanpath pipeline, where a saliency representation is first predicted and fixation sequences are then sampled from it. SaltiNet [15] introduces time-aware saliency volumes to encode temporal information before sampling scanpaths.

While another line of work follows the probabilistic approach, tSPM-Net [16], is a model that predicts complete and plausible visual exploration trajectories given a 2D input image. It uses convolutional long short-term memory recurrent networks (ConvLSTM) built on Bayesian deep learning and a novel loss function for joint spatiotemporal optimization. The model is capable of generating realistic trajectories that simulate the exploration behavior of real observers.

An alternative direction focuses on predicting future gaze based solely on past scanpaths, without relying on visual content. VPT360 [17] uses transformer-based architectures to model temporal dependencies in viewport trajectories. Although computationally efficient and effective for short-term prediction, such approaches may struggle when gaze behavior is strongly driven by salient visual events that are not encoded in the input history.

More recently, compression-inspired methods reformulate scanpath prediction as a discrete probabilistic modeling problem like ScanPath [18], by minimizing the expected code length of quantized scanpaths, the task reduces to fitting a conditional discrete probability model by maximum likelihood, and by parameterizing the output

distribution with discretized Gaussian mixture models to capture uncertainty and multimodality. Their pipeline employs a PID-controller-based sampler to produce realistic human-like scanpaths. However, the approach remains sensitive to the chosen quantization scheme and to how representative the sampled viewports are during training.

2.1.3 Datasets

Currently, there are some datasets that provides human gaze recordings of multiple observers while viewing 360° videos over a large number of videos [19, 20, 21], in which eye-tracking devices are used to capture the locations where observers fixate while wearing virtual reality headsets.

Among them is D-SAV360 [22], which is a multimodal dataset with head and eye scanpaths over 85 videos in 360° (87 participants), including estimated optical flow (generated with RAFT), saliency maps, and ambisonic audio tracks.

The VR-Eye-Tracking [23] dataset has been widely used for temporal gaze prediction in 360° videos, originally containing 208 videos across diverse dynamic scenes, with around 210k gaze samples in total.

And the Sports-360 [24] dataset that contains 108 sports videos with fast moving objects, this dataset brings together videos from five sports: basketball, parkour, BMX, skateboarding and dance.

2.2 Limitations of the state of the art and relation to this project

Across all the literature, there are various limitations that must be considered when predicting scanpaths in 360° videos, such as:

The temporal features, since videos are used instead of static images, in which the temporal dimension affects visual attention due to the presence of moving objects, where human behavior tends to prioritize areas with motion, and the order of fixations may also affect the attention, in addition to the possibility that the video camera itself may also be moving.

The fact of working in a 360° environment means that observers cannot see the entire scene at once due to the human visual range, which increases the complexity of capturing global visual attention. In addition, spherical coordinates need to be used because the images are represented using equirectangular projection in order to continue operating computationally with matrices as with 2D images.

Additionally, when predicting scanpaths, it is necessary to model the variability and uncertainty of human gaze, so that even when an observer watches the same video again, the scanpath will always be different.

This project presents a scanpath prediction framework for 360° videos that directly models the temporal evolution of visual attention. The method generates for each frame, a temporal saliency probability maps (tSPM) that capture both spatial structure and temporal dynamics in immersive content. These maps are then used to probabilistically sample coherent scanpaths, incorporating simple behavioral biases to reflect realistic viewing behavior. The proposed approach enables the generation of plausible human-like scanpaths for 360° videos without requiring new eye-tracking experiments, providing a practical tool for analyzing user attention in immersive environments.

Chapter 3

A Deep Spatio-Temporal Scanpath Prediction Model in 360° Videos

This chapter describes the implementation of the proposed deep spatio-temporal model for scanpath prediction in 360° videos. The objective is to design a deep learning model capable of modeling the temporal evolution of visual attention. This chapter focuses on the architectural design, the representation of the scanpaths so the model is able to learn to simulate them, and the optimization strategy adopted to train the model.

3.1 Architecture

The goal of the proposed architecture is to predict temporally consistent attention representations for 360° videos that can later be used to generate realistic scanpaths.

The architecture of the model is based on an encoder-decoder network, which receives as input the current frame and its corresponding optical flow. Optical flow estimates the displacement of pixels between frames, being able to represent how objects move through the scene. Each input is defined by six channels, with the frame and the optical flow each having three RGB channels. The images have a resolution of 320×240 pixels, and the input tensor has the shape $[B, 6, 320, 240]$, where B is for the batch size.

To properly handle the distortions introduced by the equirectangular projection, the architecture replace standard convolutions with spherical convolutions, where the kernel is defined as the projection onto the sphere of a small patch tangent to its surface. In this way, the sampling pattern is distorted consistently with the equirectangular image, allowing the convolution to operate on true pixel neighborhoods in spherical space. Temporal dependencies between consecutive frames are modeled using ConvLSTM modules which allow the model to learn spatial structure (Conv) and also have temporal memory (LSTM). A ConvLSTM extends the standard LSTM gating mechanism to account for spatial structure by replacing the fully connected

(matrix) multiplications with convolutional operations.

At each time step t , for a given input tensor x_t (shape $[B, C, H, W]$), previous hidden state h_{t-1} and cell state c_{t-1} , the ConvLSTM computes the input, forget and output gates and the candidate cell state using convolutional kernels.

Because the gates and states are convolutional feature maps, ConvLSTM preserves and exploits local spatial correlations while maintaining temporal memory. Stored in its hidden states and cell states keep track of evolving spatial patterns (e.g. moving salient objects or recent fixation locations) across frames. In this design, ConvLSTM is used exactly as in the original SST-Sal implementation [9], because no changes need to be made to these modules since the same information will be extracted.

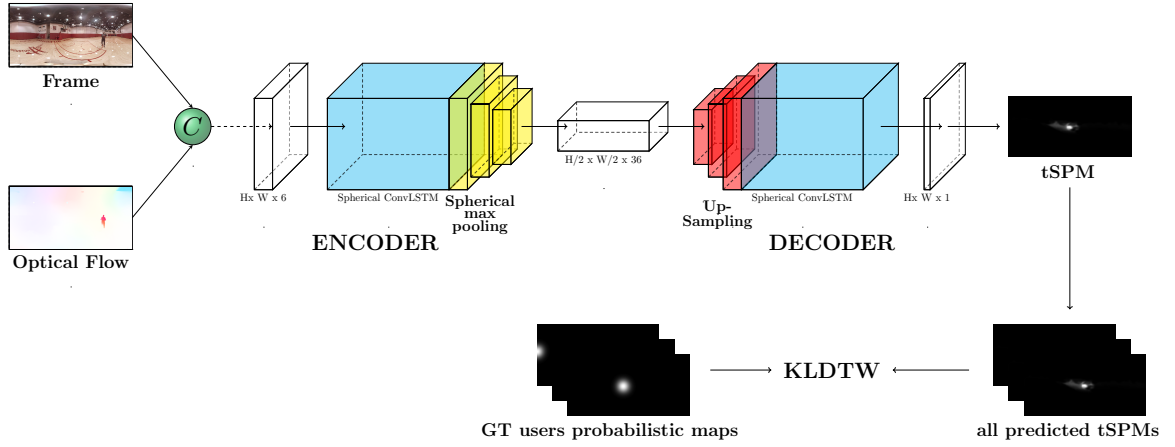


Figure 3.1: Conceptual design of the proposed spatio-temporal architecture for scanpath prediction in 360° videos, inspired by SST-Sal and tSPM-Net architectures.

The output of the network at each time step is a temporal Saliency Probability Map (tSPM), which represents a spatial probability distribution over possible gaze locations for the corresponding frame. Producing tSPMs (probabilistic maps) instead of single point predictions explicitly models the uncertainty and variability of human gaze, different observers, or the same observer at different viewings, may fixate on different locations.

As a probability map, it can represent this information in a way that allows it to be trained using loss terms (such as the KL divergence) for comparing probabilistic distributions. This makes it possible to compare this output with the ground truth fixation points, comparing it individually with each scanpath and thus learning the behavior of each trajectory separately, without having to aggregate everything onto the same saliency map. Additionally, by representing the probability distribution for each region, it facilitates separating the variability of behavior from the generation of fixation points using heuristic sampling methods. An overview of the architecture is shown in Figure 3.1.

To train the model to generate tSPMs, ground truth scanpaths are represented as Gaussian maps centered at the fixation locations where the standard deviation of each gaussian is defined as

$$\sigma = 5^\circ \cdot \frac{W}{360},$$

where W denotes the width of the image in pixels. This representation converts discrete fixation points into smooth probability distributions, making them suitable for comparison with the predicted tSPMs.

The choice of a 5° visual angle accounts for both biological and technical factors. Biologically, it covers the foveal region, which represent the effective visual span where humans process detailed information. Technically, this standard deviation compensates for spatial inaccuracies, calibration noise, and drift inherent to eye-tracking hardware. Smoothing the coordinates with a 5° kernel absorbs these errors, creating a more robust representation of human visual attention.

During training, the model is first optimized to predict saliency maps using ground truth saliency maps. Once stable spatial features are learned, the encoder is frozen and the decoder is finetuned to predict tSPMs instead of saliency maps. In this stage, the output is compared against individual scanpaths represented as Gaussian maps, enabling the model to learn temporal attention patterns at the level of individual observers.

To prevent the initialization of the ConvLSTM states from affecting the generation of the tSPM, a sliding window has been used in such a way that each window contains ten sequential frames, but only the last six frames, where the states have already stabilized, are used as valid frames to apply the loss function, so backpropagation can be performed, allowing the model to learn. Figure 3.2 diagrams the windowing scheme, that for three consecutive windows the last four frames of each window n overlap with the first four frames of window $n + 1$.

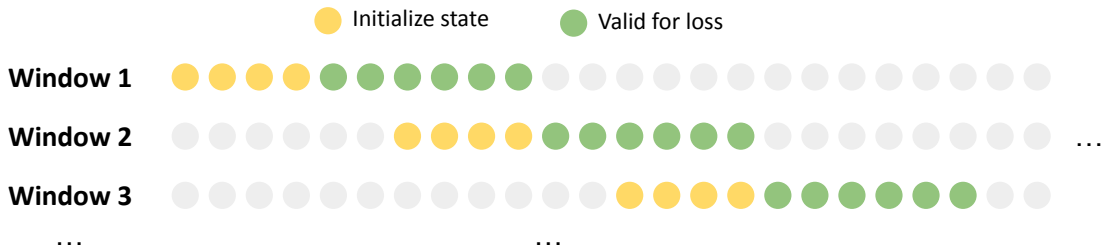


Figure 3.2: Sliding window temporal scheme. Each window processes 10 frames, the first 4 frames initialize the recurrent state (no loss performed) and the last 6 frames are compared with the loss function.

Scanpath generation is performed only during inference, where for each predicted

tSPM, a single gaze point is selected using a strategy based on the probabilities values of the tSPM. The resulting sequence of points are added sequentially to form a complete scanpath for the video. Since this sampling step is not differentiable, it is not applied during training and does not affect gradient based optimization. For more details about how the sampling method works see Chapter 4.

3.2 Loss function

To optimize the tSPM prediction model, the KLSoftDTW loss function is used (*Smoothed Dynamic Time Warping* and *Kullback-Leibler* Divergence), as used in the original work tSPM-Net [16]. In which the alignment of sequences provided by *Dynamic Time Warping* (DTW) is combined with the ability to distinguish between two probabilistic distributions provided by the *Kullback-Leibler* (KL) divergence.

Where the DTW measures the similarity between two trajectories by considering all possible alignments, without enforcing the correspondence with the time dimension. This is achieved by aligning the two trajectories so that the distance between them is minimized:

$$\text{DTW}(r, s) = \min_A \langle A, \Delta(r, s) \rangle$$

where A is a binary alignment matrix between two trajectories r and s , $\Delta(r, s) = [\delta(r_i, s_j)]_{i,j}$ is a matrix containing the distances $\delta(\cdot, \cdot)$ between each pair of points in r and s , where i and j denote the indices of the points within their respective trajectories. Finally, $\langle \cdot, \cdot \rangle$ denotes the inner product between the two matrices.

In the study tSPM-net [16], explains that the minimum function is not differentiable, so they proposed a smoothed version:

$$\text{DTW}_\gamma(r, s) = \min_A \gamma \langle A, \Delta(r, s) \rangle, \quad \gamma > 0$$

The $\min_\gamma$ function is defined as:

$$\min_\gamma(a_1, \dots, a_N) = -\gamma \log \sum_{i=1}^N \exp\left(-\frac{a_i}{\gamma}\right)$$

where $a_1, \dots, a_N$ represent the values or costs being smoothed (such as the distances between alignment steps), and N is the total number of elements in the sequence. The parameter γ adjusts the similarity between the smoothed version and the original DTW algorithm, with both being equal when $\gamma = 0$.

On the other hand, KLDiv measures the difference between two probability distributions. The Kullback-Leibler divergence (DKL) is defined as:

$$\text{DKL}(P \parallel Q) = \sum_j P(j) \log \frac{P(j)}{Q(j)}$$

where P and Q are the probability distributions to compare, and j refers to each point in the distribution [16]. In this specific case, each gaze point is represented with a gaussian map, so KLSOFTDTW can be used as a quantitative measure of how different the tSPMs are to the ground truth gaussian maps of each user.

Chapter 4

Heuristic sampling for scanpath generation

This section explains the procedure for obtaining the scanpaths, the basis of the heuristic method and the behavior of this method in different situations.

In Figure 4.1, the general processing pipeline for scanpath generation is shown:

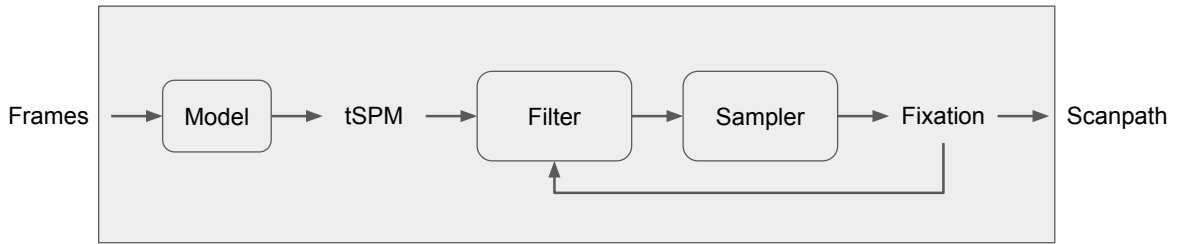


Figure 4.1: Overall pipeline of the scanpath prediction.

To obtain the tSPMs, a sliding window is used in the same way as during training, so that we only keep the predicted tSPMs that are valid, while discarding those that overlap between windows used to initialize the model state. In this way, we avoid using these repeated tSPMs when generating the scanpaths. For more details about the sliding window, see Chapter 3.

After having the tSPMs predicted by the model for the video, we can then proceed to the generation of the scanpath. Since the tSPM reflects the probabilities that the next gaze fixation point falls on each pixel of the image, a heuristic sampling method can be used, as shown in Figure 4.2 the sampling is performed probabilistically depending on the intensity of the pixels of the tSPM, which leads to different behaviors for the same tSPM.

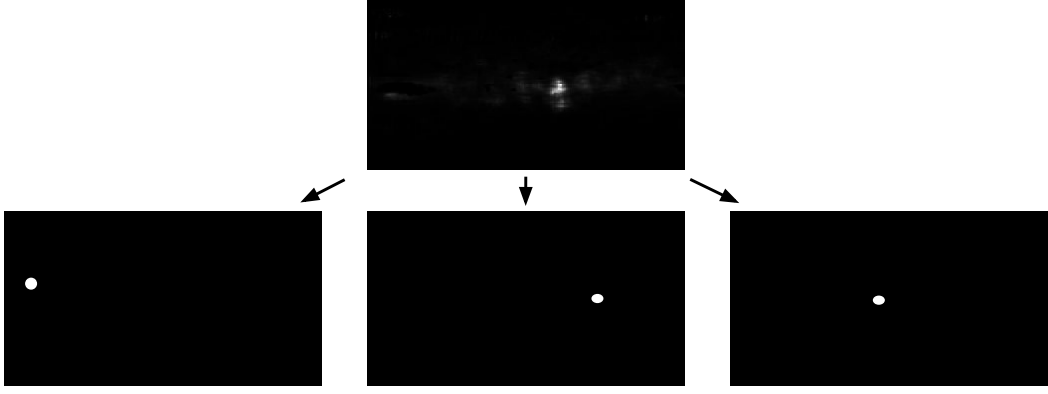


Figure 4.2: Different points are obtain when sampling probabilistically on the same tSPM.

But it would not be appropriate to sample probabilistically directly from the tSPMs, because although they reflect the spatiotemporal relationship, the next fixation will always be related to the previous one, since there is always a limit to how much a user's fixations can move within a given time interval, and that information is not contained in the tSPM. Therefore, it does not make sense to allow sampling over the entire 360° around the previous fixation.

For this purpose, a gaussian map centered on the previous fixation is generated and applied to the original tSPM as a mask to limit the maximum movement between frames. Once this mask has been multiplied with the tSPM, it must be normalized so that all pixels sum to 1, thus making it conform to a proper probabilistic distribution. Figure 4.3 shows an example of the effect of applying the Gaussian map centered on the previous fixation to the tSPM predicted by the model.



Figure 4.3: Effect of the movement gaussian on a predicted tSPM. Left: original tSPM; Center: Gaussian centered at the previous fixation. Right: filtered and renormalized tSPM used for sampling.

To apply to the tSPM the mask defined as the gaussian map centered on the previous fixation, we need to know the value of the standard deviation of the gaussian distribution, since it determines the maximum range of movement. This range is defined by several factors, such as user movement while viewing the video, as well as

the time difference between the fixation points we want to sample, which in this case is 400 ms between each tSPM.

Therefore, if small values (e.g., 5°) are used for the standard deviation of the gaussian, a very static behavior would be obtained, which would not allow movement toward nearby regions of higher attention at a normal speed. On the other hand, if a very high value (e.g., 80°) were used, exploration of the scene toward remote locations with respect to the previous fixation point would be encouraged, generating very unnatural and erratic movement in cases where the tSPM represents two attention regions that are very far apart from each other.

To adjust the value of the standard deviation of the gaussian map, an analysis was performed on the VR-Eye-Tracking dataset [23]. Given the fixation points of all users, the spherical distance in degrees between one point and the next was calculated, adjusted to the time interval (400 ms) at which the model operates. Once these distances are obtained, to prevent outliers from affecting the result, the interquartile range ($\text{IQR} = (Q_3 - Q_1)$) is used to calculate the standard deviation, with the formula being as follows:

$$\sigma_{\text{deg}} = \text{IQR} \cdot 1.3$$

It would not be correct to use the maximum motion between frames as the standard deviation of the Gaussian, since this would be treated as an outlier that would negatively affect the results, as it would not resemble the real behavior. Therefore, a data point is considered an outlier when it exceeds the IQR value (which corresponds to 50% of all the data) by a multiplier. With all the data, the IQR represents 50% of the data, but to remove outliers, a multiplier is used to set the threshold for considering data as outliers, both high and low. In this case, however, we are interested in retaining the maximum range of movement, so we take the upper limit using Q_3 , ensuring that 100% of the data falls below this value. Therefore, to consider a correct gaussian distribution, it can be assumed that this value corresponds to $2\sigma_{\text{deg}}$.

In summary, to predict a scanpath, a video is passed through the model to obtain the tSPMs, to which a mask in the form of a Gaussian map centered on the previous fixation is applied. After normalizing the result, probabilistic sampling is used to obtain the fixation point for the current frame (which will be used in the mask for the next frame). Once the entire video has been processed, all the points are concatenated to obtain the predicted scanpath.

Chapter 5

Evaluation and Results

The objective of the evaluation is to analyze how good the method proposed in this project is, in order to determine whether it is capable of generating realistic scanpaths that resemble human behavior.

To this end, the methodology for comparing this method with baselines and with the state of the art is explained below, which will be analyzed both qualitatively and quantitatively.

For each video, multiple scanpaths are generated and individually compared with each of the ground truth scanpaths using metrics to measure the distance between different trajectories. Once the results are obtained, the mean and standard deviation of the comparisons between a predicted scanpath and all ground truth scanpaths are calculated. Then, the results of all predicted scanpaths for a video are averaged, and finally, the results are averaged again across all videos.

5.1 Metrics

For this evaluation, four metrics are used to measure the distance between different trajectories, which are commonly employed in studies dealing with scanpaths and spherical coordinates: Dynamic Time Warping (DTW), Levenshtein distance (LEV), Euclidean distance (EUC), and Fréchet distance.

With these metrics, it is possible to analyze whether the methods resemble the spatio-temporal characteristics of real scanpaths. There are other metrics commonly used in this field, but they are not suitable for this specific context, such as determinism or recurrence, since they lose their meaning when applied to the temporal characteristics of videos.

For example, recurrence, used for scanpaths in static images, serves to measure whether the trajectory tends to return to the same area, representing an object of interest. However, in the case of videos, this does not necessarily mean that the viewer

is returning to the same object, since it may be in motion. For this reason, only the metrics that better measure when attention in the scene is focused on moving objects are used.

5.1.1 Euclidean distance (EUC)

Euclidean distance measures the spatial error between two points, so to compare two trajectories, the distance between points must be calculated and averaged. In this case, spherical coordinates are used to account for the equirectangular projection, meaning that temporal alignment between the trajectories is not considered. The Euclidean distance is computed as:

$$\text{EUC}(R, S) = \frac{1}{K} \sum_{k=1}^K \|r_{i_k} - s_{j_k}\|_2$$

where $\|\cdot\|_2$ denotes the Euclidean norm in the spherical coordinate representation.

5.1.2 Dynamic time warping (DTW)

Dynamic time warping (DTW) finds the optimal temporal alignment between two scanpaths and produces a sequence of matched index pairs, so that it is able to align two different sequences with the same pattern occurring at different times. This metric is explained in depth in Chapter 3.

5.1.3 Levenshtein distance (LEV)

Levenshtein distance, also named edit distance, measures the minimum number of changes that would need to be applied to a scanpath to transform it into another using edit operations (insertions, deletions, substitutions).

5.1.4 Fréchet distance

Fréchet distance [25] quantifies the similarity of two trajectories while respecting the ordering of points along them. For example, it is the minimum leash length required for a person and a dog to traverse the two trajectories from start to end without backtracking.

Fréchet distance is stricter than the mean Euclidean distance and is sensitive to large local deviations, since it emphasizes the global shape and ordering of trajectories.

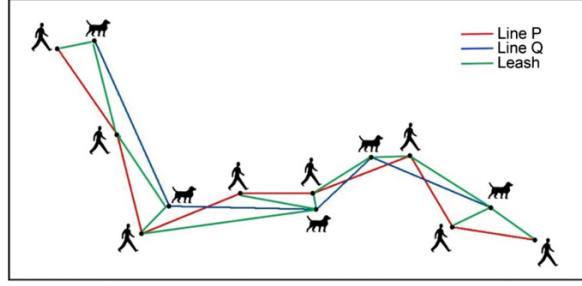


Figure 5.1: Illustration of two scanpaths, and the Fréchet leash between them [25].

5.2 Dataset

Two different datasets are used in the evaluation:

The VR-Eye-Tracking [23] dataset that provides a large eye-tracking collection that contains 208 dynamic 360° video clips (20s to 70s each) with eye fixation recordings from multiple subjects and around 210k gaze samples in total, the original study reports sessions with on the order of 25-33 viewers and provides fixation points together with head positions [23].

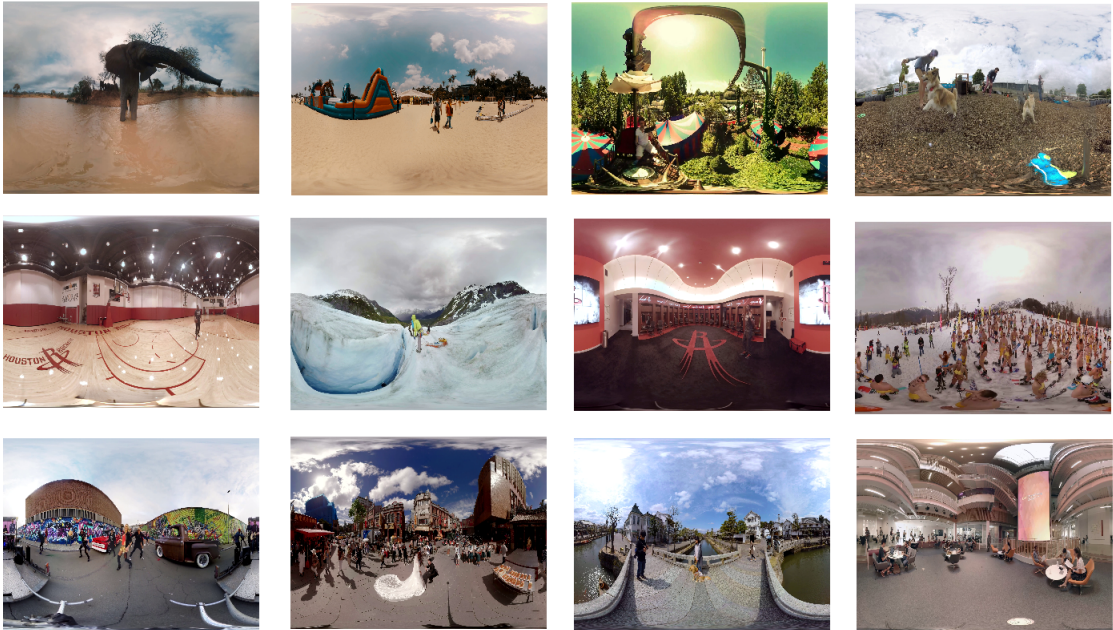


Figure 5.2: Representative frames of some of the Vr-Eye-Tracking dataset videos.

And the Sports-360 [24] dataset, is a large collection of 360° videos on sports themes. The dataset have 108 panoramic videos across five sports domains (basketball, parkour, BMX, skateboarding and dance), and includes human ground truth fixations agents[24].

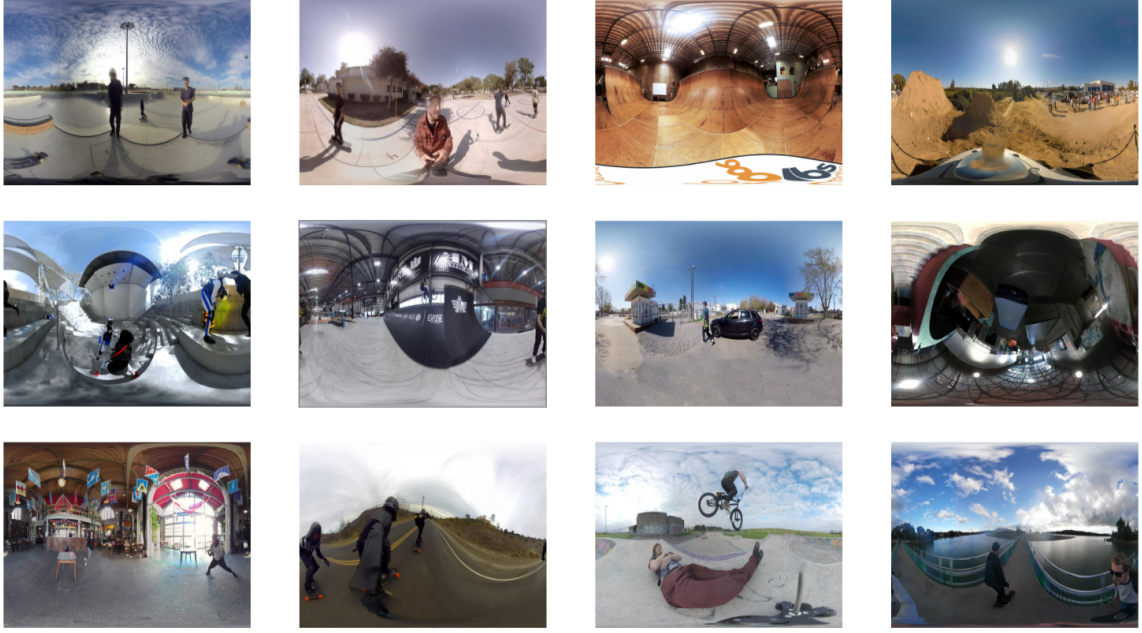


Figure 5.3: Representative frames of some of the Sports-360 dataset videos.

For this project the VR-Eye-Tracking was used for training, validation and testing after an explicit filtering step: videos with significant camera motion and clips that incentive random exploration, were discarded leaving 148 usable videos from the original 208. The split is performed at the video level to prevent leakage: 20% of the filtered videos were held out as a test set, while the remaining 80% were split internally (80% training, 20% validation). The Sports-360 collection was used exclusively as an additional test set: only Sports-360 videos that meet the same filtering criteria were used for testing to assess cross-dataset generalization.

5.3 Baselines and methods

In this evaluation, different methods and baselines are compared, such as the *human baseline*, *pure random baseline*, *clever random baseline*, the model proposed in this project, the adapted SST-Sal, and the current state of the art, SPath.

The *human baseline* consists of comparing the ground truth scanpaths with themselves, in order to determine the human behavior that the evaluated method aims to match in order to be considered realistic and well generalized.

The *pure random baseline* consists of generating completely random fixation points without taking into account any bias or limitation. It is used to represent a worst case scenario, considering that at a minimum any method should outperform its results.

The *clever random baseline* is a heuristic method that consists of generating fixation points with some typical biases of human behavior. It is used to assess what results

can be obtained by a simpler method that does not rely at all on information from the video. More specifically, equator bias is used, which, as indicated in D-SAV360 [22], shows that the vast majority of fixations are located near the horizon, as this is the area where humans naturally tend to focus their attention. This bias can be represented in a probabilistic map with parameters $\mu = 91.8^\circ$ and $\beta = 23.19^\circ$ [22], as explained in the mentioned study. In addition, the maximum movement between fixations is also limited by generating a gaussian map centered on the previous fixation, as explained in more detail in Chapter 4.

For the proposed model, the combination of the deep learning model and the heuristic method is used, which can be reviewed in Chapter 3 and Chapter 4, respectively, where their operation is explained in detail.

In the case of the adaptation of SST-Sal [9], it consists of the same pipeline as the method proposed in this project, but replacing the tSPM generation model with the SST-Sal model, which generates saliency maps, while the rest of the pipeline remains the same. This method is used as a comparison between using saliency maps and tSPMs to determine which one better leverages the heuristic method.

Finally, Spath[18] is used as the available state of the art model for scanpath generation in 360° videos, in order to determine whether the method of this project can match or even improve upon the state of the art.

5.4 Results

The results obtained using the different metrics on the two datasets can be seen in Table 5.1 (for the VR-Eye-Tracking dataset) and Table 5.2 (for the Sports-360 dataset), where the same metrics and methods are presented for easier comparison, where lower results indicate better performance.

Metric	DTW ↓		LEV ↓		EUC ↓		Frechet ↓	
Human BL	1656.5967	(704.4053)	97.9703	(25.7116)	293.5168	(101.5599)	54.0204	(17.5237)
Pure Random	3478.8396	(225.1127)	151.1585	(4.0119)	422.9451	(28.8518)	64.0856	(8.2049)
Clever Random	2397.4004	(350.4870)	131.1533	(6.5948)	344.6316	(42.9701)	52.8536	(8.4806)
SSt-Sal	<u>1745.6083</u>	(352.8610)	108.5304	(10.5760)	<u>277.3693</u>	(47.9744)	<u>47.6290</u>	(9.3403)
Spath	2607.4216	(598.3377)	133.3322	(11.3299)	355.6595	(62.6937)	68.0687	(9.3458)
Our model	1726.4520	(387.4023)	<u>110.4364</u>	(9.9423)	270.4801	(51.5799)	46.6286	(9.3760)

Table 5.1: Quantitative result on the VR-Eye-Tracking test set.

Metric	DTW ↓		LEV ↓		EUC ↓		Frechet ↓	
Human BL	1106.8277	(540.6844)	74.2216	(24.5022)	217.4360	(84.8540)	42.2446	(16.0864)
Pure Random	2794.8171	(952.7877)	114.8700	(38.3836)	433.6357	(146.2761)	66.0347	(25.5679)
Clever Random	2024.2755	(485.5569)	107.9574	(22.3994)	347.8065	(79.7242)	55.1050	(17.4536)
SSt-Sal	1448.4125	(542.8502)	82.2520	(27.8120)	282.3164	(101.9264)	<u>49.0322</u>	(20.5529)
Spath	1556.4308	(558.8159)	91.0877	(21.3382)	271.8155	(86.7814)	48.1782	(18.8200)
Our model	<u>1464.8405</u>	(568.4520)	<u>84.2200</u>	(28.6588)	<u>277.4737</u>	(101.6969)	50.2680	(21.3149)

Table 5.2: Quantitative result on the Sports-360 dataset.

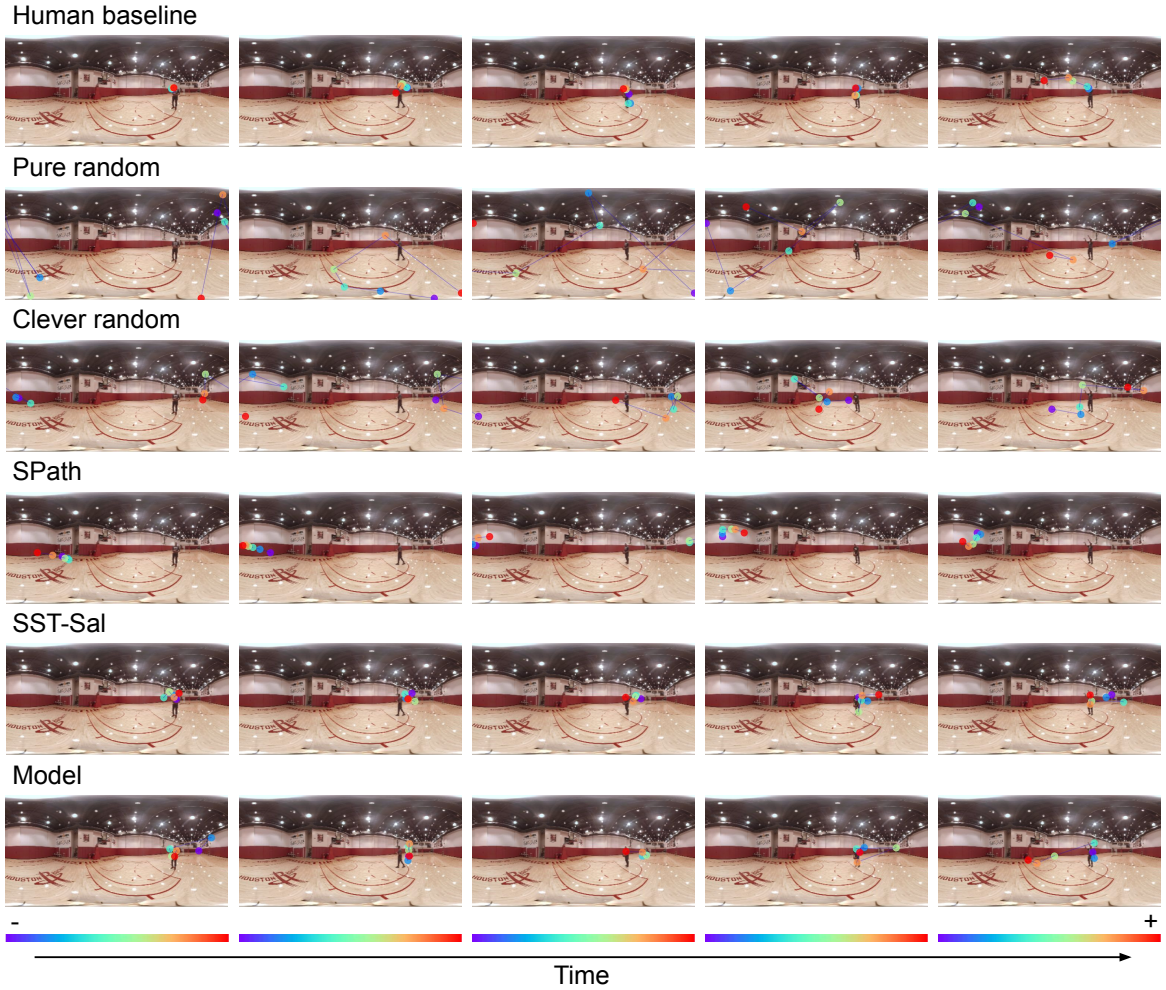


Figure 5.4: Qualitative comparison of scanpaths across methods, the horizontal axis represents time, where the most recent point (red color) of a frame corresponds to the oldest point (purple color) of the following frame.

In addition, visual representations of the scanpaths have been created to compare them graphically in cases where the metrics are unable to measure similarity with human behavior. In Figure 5.4, the evolution of the scanpaths for the different methods used on the same example video can be seen, while in Figure 5.5 and Figure 5.6, the

different scanpaths generated by the model can be observed in various videos of the VR-Eye-Tracking and Sports-360 datasets, respectively, can be observed.

In the following links, both the results obtained on **Drive**, and the framework source code on **Github** are available to promote future extensions or evolution of this work.

5.4.1 Quantitative analysis

Overall, the proposed method achieves the best performance across the majority of metrics and datasets. On the VR-Eye-Tracking dataset, it consistently outperforms the state of the art method and the adaptation of the SST-Sal model [9], on which this project is based, in which the same sampling method as in Chapter 4 is followed. This means that the use of tSPMs over saliency maps increases the ability to obtain spatiotemporal information about the temporal dependency of a scanpath, which saliency maps are unable to capture.

Looking at the results, it can be observed that depending on the dataset, SPath [18] shows worse performance, to the point that in the VR-Eye-Tracking dataset, the Clever Random method obtains better results, despite only using the equator bias to define the horizon as the most common area where an observer places their gaze, and a filter to limit movement speed so that it resembles human behavior. SPath is likely obtaining worse results on this specific dataset due to the length of the sequences, since the model works better on clips lasting up to 30 seconds, whereas the VR-Eye-Tracking [23] dataset contains clips lasting between 20 and 70 seconds, making the predictions less consistent due to limitations in long-term scanpath prediction.

Additionally, as can be seen in Figure 5.6, results are also obtained that improve all the metrics compared to all the different methods, even outperforming the current state of the art. This indicates that the method proposed in this project is capable of generating realistic scanpaths, and furthermore, since this dataset has only been used in the testing stage, the method can be considered robust and capable of generalizing to all kinds of situations.

5.4.2 Qualitative analysis

Because scanpaths exhibit so much variability and uncertainty, it is very difficult for metrics to serve as definitive proof of good performance.

For this reason, performing quantitative analysis together with qualitative analysis, in which scanpaths are plotted visually, can help with the comparison of different methods. Since, particularly in the case of human behavior, it is much easier

to recognize visual patterns as humans, even when we cannot clearly explain why something does not seem realistic. Whereas computationally it is necessary to perform complex algorithms to determine whether a scanpath is realistic.

For example, in the case of the SPath method, it has several drawbacks, such as the fact that it always requires ground-truth scanpaths to be provided in order to generate results. It directly uses a few initial points from a ground-truth scanpath and then starts generating its own points. If it happens that the ground truth at the beginning is looking at an abnormal region, this ends up conditioning the trajectory, making it look unrealistic. In addition, the movement it generates between points along the trajectory is very robotic, as can be seen in Figure 5.4, where it moves in a single direction little by little without being affected by the situation of the scenario. In fact, it can be observed that it lacks movement toward the regions of visual attention (which are mainly along the trajectory of the human baseline).

Regarding the random methods, it can be observed that they do not follow the visual attention of the GT scanpaths at all, appearing as if they are not looking at anything in particular, but rather moving constantly without maintaining attention on any location. However, it can be noticed that the clever random baseline moves mainly along the horizon, appearing much more realistic than being completely random, as in the pure random baseline.

For the adaptation of SST-Sal and the model proposed in this project, they do not appear to differ much visually from each other, but it can be noted that they tend to move through areas where there are moving objects or locations that would attract attention when passing by. All this is due to being based on probabilistic maps (saliency maps and tSPMs) that determine the most probable areas to which they will move.

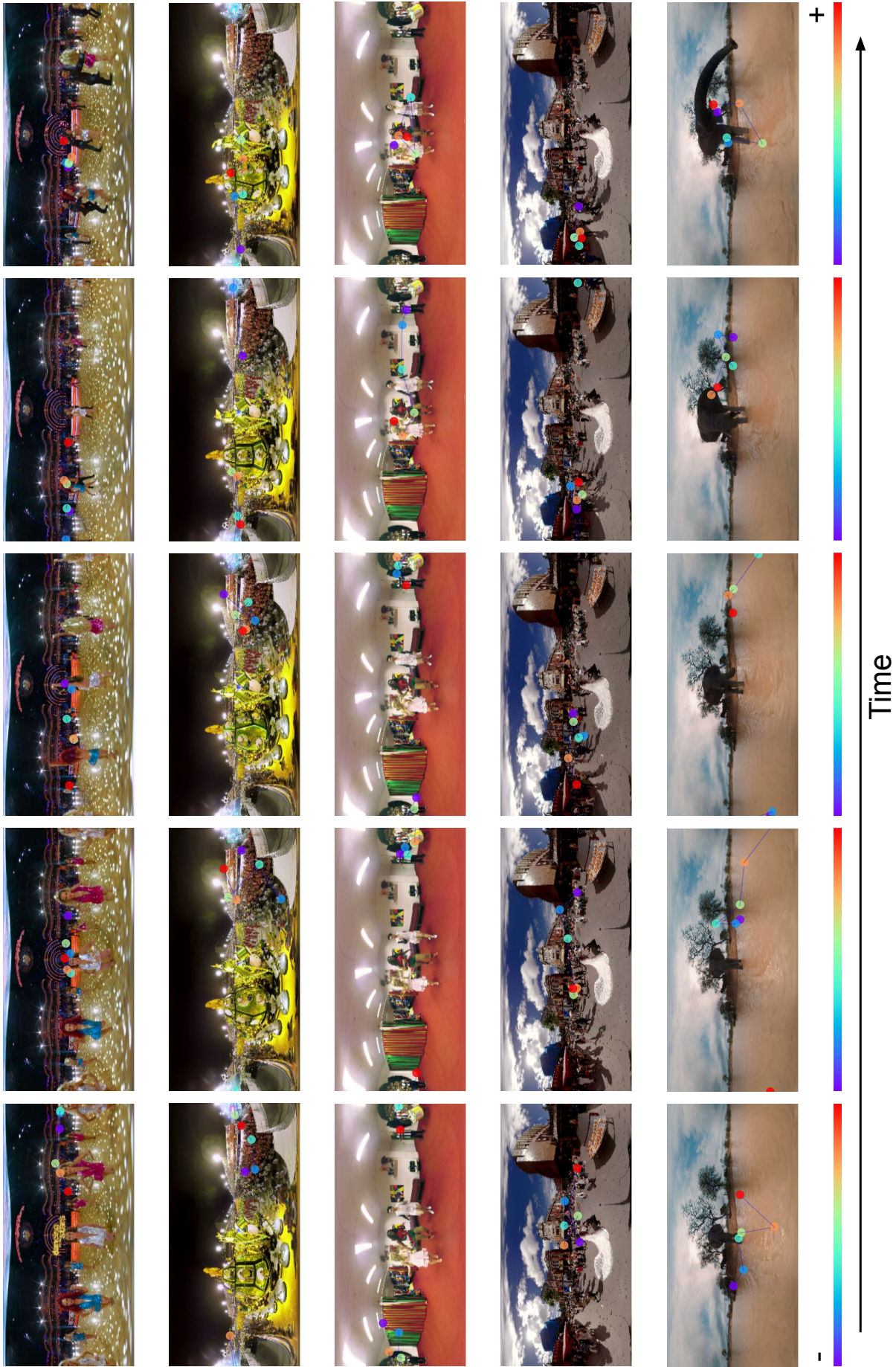


Figure 5.5: Qualitative comparison of predicted scanpaths by the proposed model in different videos of the VR-Eye-Tracking dataset, the horizontal axis represents time, where the most recent point (red color) of a frame corresponds to the oldest point (purple color) of the following frame.

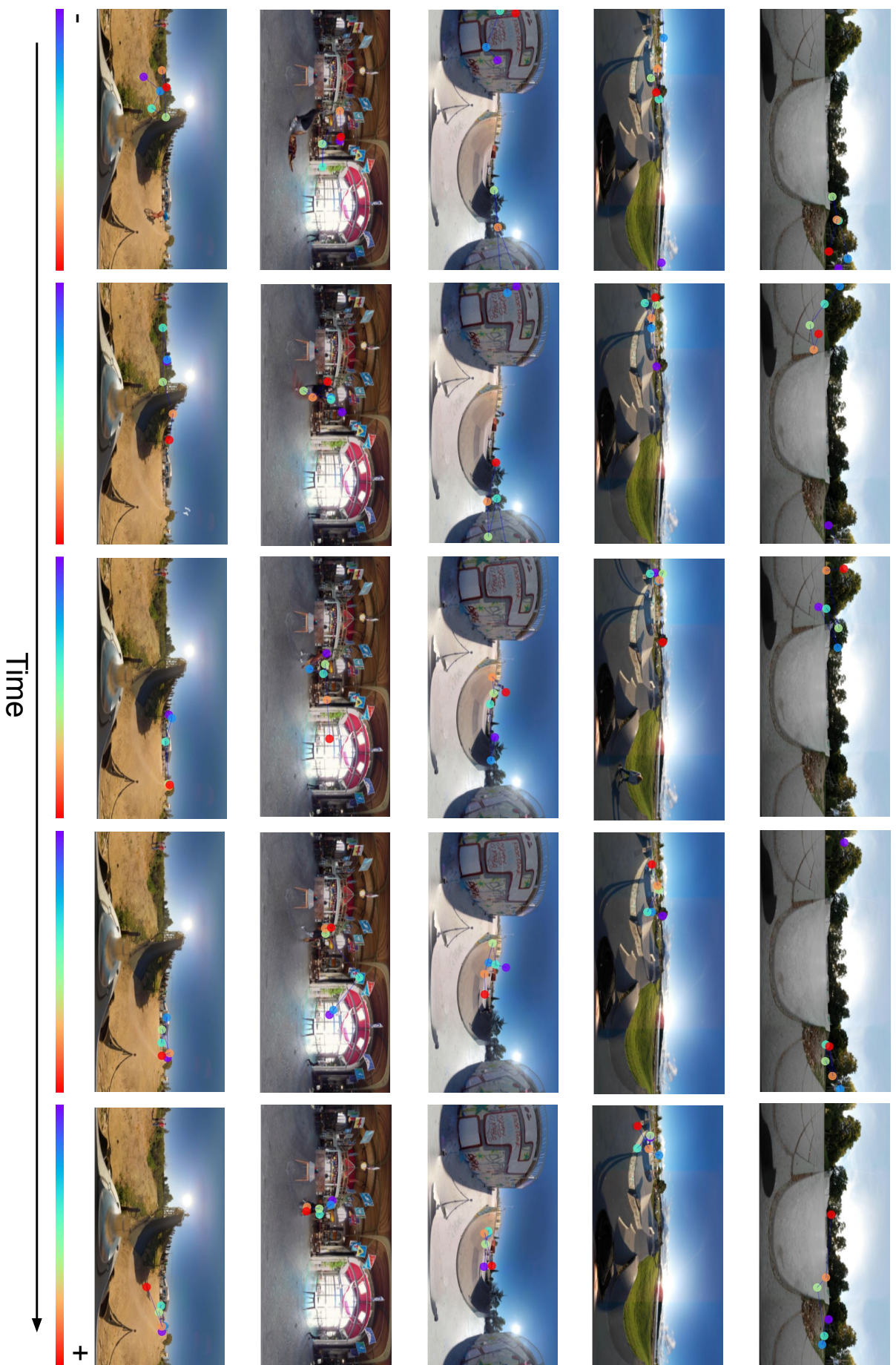


Figure 5.6: Qualitative comparison of predicted scampaths by the proposed model in different videos of the Sports-360 dataset, the horizontal axis represents time, where the most recent point (red color) of a frame corresponds to the oldest point (purple color) of the following frame.

Chapter 6

Conclusions

The objective of this project is to predict scanpaths in 360° , using a method that employs a deep learning model to capture spatiotemporal dependencies by generating tSPMs, and combines this with a heuristic sampling method over these probability maps to generate realistic scanpaths.

The evaluation on the VR-Eye-Tracking and Sports-360 datasets show that the proposed approach produces scanpaths that are superior to different baselines and state of the art methods. In particular, when attempting to predict scanpaths, it is necessary to take into account the ways of representing visual attention in order to obtain information about the areas toward which the gaze is directed. In addition, the proposed method exemplifies how the use of tSPMs over saliency maps improves the ability to capture temporal dependencies.

6.1 Limitations and future Work

Although this method achieves results that are above the state of the art, it still has limitations that can be improved:

First, the fact that the model needs to use sliding windows to process long sequences imposes a restriction on its ability to capture long-range temporal dependencies, where other types of architecture could improve temporal dependency modeling.

Additionally, the model could improve the capability to predict tSPMs by including information about the trajectory followed by the scanpath, thus directly incorporating spatial dependency as input to the model, in a way that the extraction of this input is differentiable, so that the backward pass of the model can be performed in the training.

Since it is working with videos, it is important to consider that audio is also a source of information that contributes to attention, as there are sounds associated with objects and events. Furthermore, if the sound is directional, it can be used as an attention cue to guide gaze, which is a resource that is not taken into account in this project.

Chapter 7

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